

Formalização do Modelo v2

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1 Environment

Definition 1 (Institutional Design Game). *The game $\Gamma(N, p, r, \alpha, \beta, c)$ has the following primitives.*

- *There are $N \geq 3$ players: one hegemon H and $n = N - 1$ weak states $W_1, \dots, W_n$.*
- *Nature draws $\theta \in \{0, 1\}$ with prior $\Pr(\theta = 1) = p \in (0, 1)$.*
- *The state determines the value of multilateral cooperation: $V(0) = 1$ and $V(1) = r > 1$.*
- *The hegemon observes θ ; the weak states do not.*
- *If bargaining fails (disagreement):*
 - *Each weak state receives 0.*
 - *The hegemon receives $\alpha V(\theta)$, where $\alpha \in (0, 1/r)$, representing bilateral alternatives proportional to the value of cooperation.*
- *The common discount factor is $\beta \in (0, 1)$.*
- *Each weak state pays an entry cost $c > 0$ to join the institution.*

Let

$$V_e(\mu) \equiv 1 + \mu(r - 1)$$

denote the expected value of cooperation at posterior belief μ .

Remark 1 (Normalization). *The assumption $d_W = 0$ is without loss of generality. It requires only that (i) weak states' outside options are strictly lower than the hegemon's, and (ii) outside options are symmetric across weak states. Given these conditions, we redefine the pie as the surplus above the weak states' common outside option, which normalizes their disagreement payoff to zero.*

Remark 2 (Bilateral alternatives). *The hegemon's disagreement payoff $\alpha V(\theta)$ captures bilateral alternatives that are proportional to the value of multilateral cooperation. The same factors that make multilateral cooperation valuable (favorable trade environment, complementary economies) also improve bilateral alternatives. The parameter $\alpha < 1/r$ ensures that bilateral alternatives are strictly dominated by multilateral cooperation for all θ .*

2 Institutional Packages

Definition 2 (Institutional Packages). *The analysis compares two institutional packages that differ only in the voting rule. Both packages have symmetric proposal rights (random proposer with probability $1/N$ each).*

- **Package M (Majority):** decisions pass with strict majority support ($q = \lfloor N/2 \rfloor + 1$ votes).
- **Package U (Unanimity/Consensus):** decisions require unanimous support ($q = N$ votes).

Remark 3 (Why not agenda control?). *By agenda control we mean here monopoly over proposal rights (exclusive proposer). Under either voting rule, if the hegemon were the exclusive proposer and $d_W = 0$, weak states' expected bargaining payoff would be zero. No weak state would pay the entry cost $c > 0$, and the institution would never form. Monopoly proposal power is therefore self-defeating. The relevant institutional choice is between voting rules, holding proposal rights symmetric. Other forms of agenda influence—such as shaping the set of feasible proposals—are discussed in the extension.*

3 Timing

The game has three stages.

1. **Stage 0 (Institutional choice).** The hegemon chooses the voting rule $R \in \{M, U\}$.
2. **Stage 1 (Information and entry).** Nature draws θ . The hegemon observes θ and commits to a public signal structure $\pi : \{0, 1\} \rightarrow \Delta(S)$. Weak states observe the realized signal $s \in S$, update to posterior $\mu = \Pr(\theta = 1 \mid s)$, and simultaneously decide whether to enter (paying cost c).
3. **Stage 2 (Bargaining).** If the institution forms, members play a two-round Baron-Ferejohn bargaining game over a pie of size $V(\theta)$:
 - **Round 1:** a proposer is selected uniformly at random (probability $1/N$). The proposer

offers a division. All members vote according to rule R . If accepted, the pie is divided as proposed. If rejected, the game proceeds to Round 2.

- **Round 2 (terminal):** a new proposer is selected uniformly at random. The proposer offers a division. All members vote. If accepted, the pie is divided. If rejected, each player receives their disagreement payoff (0 for each W , $\alpha V(\theta)$ for H).

4 Solution Concept

The solution concept is Perfect Bayesian Equilibrium (PBE). Strategies are sequentially rational given beliefs, and beliefs are updated by Bayes' rule on the equilibrium path. Off-path beliefs satisfy sequential rationality.

5 Game Tree

The game tree is presented for the unanimity subgame (Package U). Under Package M (majority), the structure is identical except that W proposers exclude H from the coalition, eliminating the screening problem.

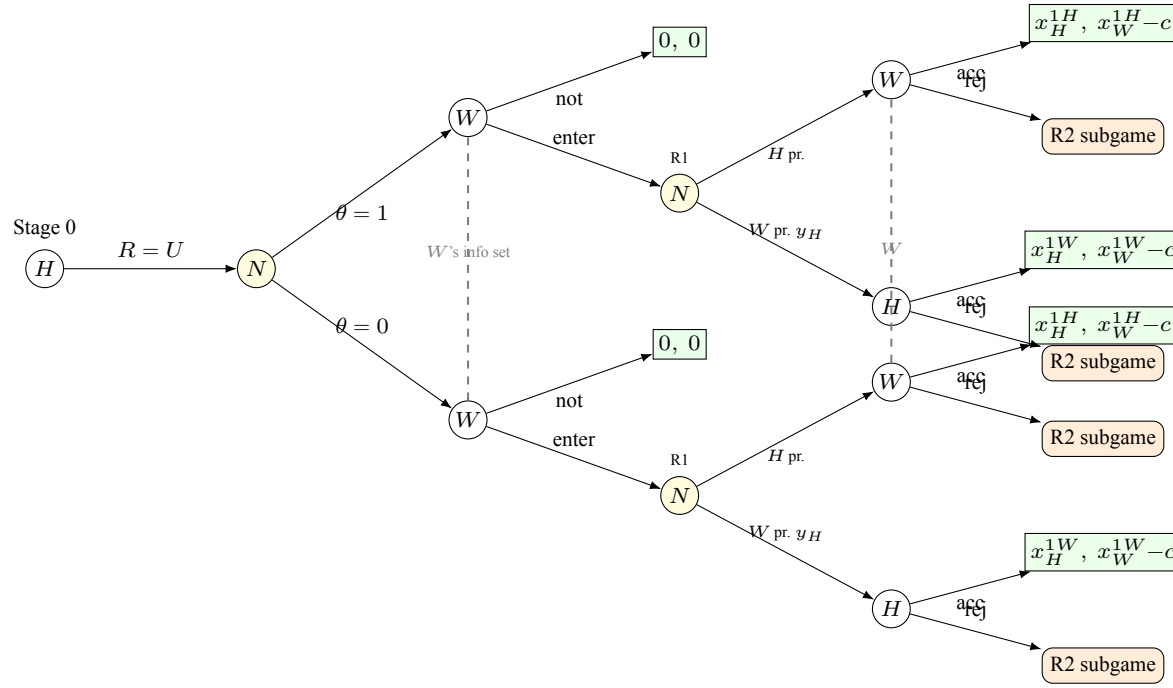


Figure 1: Main game tree under Package U (unanimity): Stage 0 through Round 1. H chooses R ; Nature draws θ ; H sends signal (not shown); W forms posterior μ and decides entry (paying c only if entering). In R1, Nature selects proposer ($1/N$). When H proposes, W votes to accept or reject. When W proposes, W offers y_H to H , who accepts or rejects. Any rejection in R1 leads to the R2 subgame (Figure 2). Dashed lines: W 's information sets (W does not observe θ). Under Package M , W proposers exclude H , eliminating the screening branch.

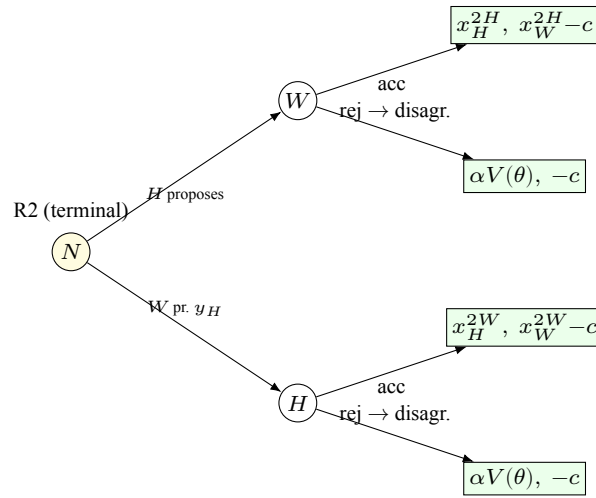


Figure 2: R2 subgame (terminal round) under unanimity. This subgame is reached only when W plays aggressive in R1 and $\theta = 1$ rejects. Therefore, W **knows** $\theta = 1$ (complete information — no screening in R2). Nature selects proposer. When W proposes, W offers αr to H (exact reservation of $\theta = 1$). When H proposes, H offers 0 to W . Rejection leads to disagreement: $\alpha V(1)$ for H , $-c$ for W .

6 Equilibrium Analysis (to be developed)

6.1 Terminal Round (R2) under Unanimity

When H proposes (prob $1/N$): offers 0 to each W ($d_W = 0$), all W 's accept (indifferent: $0 \geq 0$). H keeps $V(\theta)$. This is the same for both types and does not involve screening. When W decides to accept or reject, W also doesn't know θ , but since the offer is 0 and $d_W = 0$, W is indifferent and accepts by convention.

When W_j proposes (prob $1/N$): offers 0 to other W 's (who accept by the same indifference argument). W_j must decide what to offer H . This is the screening subgame, which we now analyze via the normal form.

6.1.1 R2 Screening Subgame: Normal Form

Players: W (proposer, uninformed) and H (responder, type $\theta \in \{0, 1\}$).

Strategy spaces. W chooses an offer y_H . By dominance, only two offers are relevant:

- **Aggressive (A):** $y_H = \alpha = \alpha V(0)$
- **Conservative (C):** $y_H = \alpha r = \alpha V(1)$

Any $y_H < \alpha$ is rejected by both types; any $y_H \in (\alpha, \alpha r)$ is accepted by $\theta = 0$ and rejected by $\theta = 1$ (same as A but costlier); any $y_H > \alpha r$ is accepted by both (same as C but costlier).

H of type θ chooses accept/reject for each possible offer. Writing H 's strategy as $(a_\theta^A, a_\theta^C) \in \{a, r\}^2$:

Dominance elimination:

- **Type $\theta = 1$ facing A :** reject strictly dominates ($\alpha r > \alpha$). So $a_1^A = r$ always.
- **Type $\theta = 0$ facing C :** accept strictly dominates ($\alpha r > \alpha$). So $a_0^C = a$ always.

After elimination, the remaining choices are:

- Type $\theta = 0$: $a_0^A \in \{a, r\}$ (accept or reject the aggressive offer)
- Type $\theta = 1$: $a_1^C \in \{a, r\}$ (accept or reject the conservative offer)

H 's reduced strategy set: $(a_0^A, a_1^C) \in \{a, r\}^2$, giving 4 strategies. W 's strategy set: $\{A, C\}$.

Key observation: H is indifferent at the non-dominated margins:

- Type $\theta = 0$ facing A : accept gives α ; reject gives $\alpha V(0) = \alpha$. **Indifferent.**

- Type $\theta = 1$ facing C : accept gives αr ; reject gives $\alpha V(1) = \alpha r$. **Indifferent.**

Therefore H 's payoff does not depend on H 's own strategy:

$$u_H(\theta = 0) = \alpha \text{ if } W \text{ plays } A, \quad \alpha r \text{ if } W \text{ plays } C \quad (1)$$

$$u_H(\theta = 1) = \alpha r \text{ regardless of } W\text{'s action} \quad (2)$$

W 's payoff matrix (expected over θ given beliefs $\mu = \Pr(\theta = 1)$):

	(a, a)	(a, r)	(r, a)	(r, r)
A	$(1 - \mu)(1 - \alpha)$	$(1 - \mu)(1 - \alpha)$	0	0
C	$V_e(\mu) - \alpha r$	$(1 - \mu)(1 - \alpha r)$	$V_e(\mu) - \alpha r$	$(1 - \mu)(1 - \alpha r)$

Columns are H 's strategies (a_0^A, a_1^C) . Note: H 's payoff is the same across all columns (indifferent).

BNE analysis. Since H is indifferent, any strategy is a best response for H . The equilibrium is determined by W 's best response:

- Under (a, a) : W compares $(1 - \mu)(1 - \alpha)$ vs $V_e(\mu) - \alpha r$. Cutoff at μ_s .
- Under (a, r) : W compares $(1 - \mu)(1 - \alpha)$ vs $(1 - \mu)(1 - \alpha r)$. Since $1 - \alpha > 1 - \alpha r$, A dominates.
- Under (r, a) : W compares 0 vs $V_e(\mu) - \alpha r > 0$. C dominates.
- Under (r, r) : W compares 0 vs $(1 - \mu)(1 - \alpha r) > 0$. C dominates.

Multiple BNE exist (because H is indifferent). The standard convention in bargaining games is: **accept when indifferent**. This selects (a, a) : type $\theta = 0$ accepts A , type $\theta = 1$ accepts C .

Under (a, a) , W 's screening cutoff is:

$$(1 - \mu)(1 - \alpha) = V_e(\mu) - \alpha r \implies \mu_s = \frac{\alpha(r - 1)}{r - \alpha} \quad (3)$$

W plays A if $\mu < \mu_s$, C if $\mu > \mu_s$.

6.1.2 R2 Continuation Values (combining H-proposes and W-proposes)

Averaging over proposer identity (prob $1/N$ each):

$V_W^{R2}(\mu)$:

$$\mu < \mu_s : \frac{(1 - \mu)(1 - \alpha)}{N} \quad (4)$$

$$\mu > \mu_s : \frac{V_e(\mu) - \alpha r}{N} \quad (5)$$

Continuity check at μ_s : at $\mu_s = \alpha(r - 1)/(r - \alpha)$, we have $1 - \mu_s = r(1 - \alpha)/(r - \alpha)$, so $(1 - \mu_s)(1 - \alpha) = r(1 - \alpha)^2/(r - \alpha)$. And $V_e(\mu_s) - \alpha r = 1 + \mu_s(r - 1) - \alpha r = 1 + \alpha(r - 1)^2/(r - \alpha) - \alpha r$. Simplifying: $= [r - \alpha + \alpha(r - 1)^2 - \alpha r(r - \alpha)]/(r - \alpha) = [r - \alpha + \alpha r^2 - 2\alpha r + \alpha - \alpha r^2 + \alpha^2 r]/(r - \alpha) = [r - 2\alpha r + \alpha^2 r]/(r - \alpha) = r(1 - \alpha)^2/(r - \alpha)$. Equal. ✓

$V_H^{R2}(\theta, \mu)$:

$$\theta = 1 : \frac{r[1 + (N - 1)\alpha]}{N} \quad (\text{constant in } \mu) \quad (6)$$

$$\theta = 0, \mu < \mu_s : \frac{1 + (N - 1)\alpha}{N} \quad (7)$$

$$\theta = 0, \mu > \mu_s : \frac{1 + (N - 1)\alpha r}{N} \quad (\text{overpaid}) \quad (8)$$

Jump in $E_\theta[V_H^{R2}(\mu)]$ at cutoff μ_s :

$$\text{Jump} = \frac{(N - 1)\alpha(r - 1)(1 - \mu_s)}{N} > 0 \quad (9)$$

6.1.3 Source of the difference between unanimity and majority

Proposition 1 (Overpayment under unanimity). *Under unanimity on the conservative branch ($\mu > \mu_s$), type $\theta = 0$ receives strictly more than under majority:*

$$V_H^{R2}(\theta = 0, U, \mu > \mu_s) = \frac{1 + (N - 1)\alpha r}{N} > \frac{1 + (N - 1)\alpha}{N} = V_H^{R2}(\theta = 0, M).$$

On the aggressive branch ($\mu < \mu_s$), payoffs are identical across rules. Type $\theta = 1$ receives the same under both rules for all μ .

Proof. Under majority, W excludes H ; H receives the disagreement payoff $\alpha V(\theta)$, which is type-specific: α for $\theta = 0$, αr for $\theta = 1$. Under unanimity, W must include H but does not know θ . On the conservative branch, W offers αr to *both* types. Type $\theta = 1$ receives αr (same as majority).

Type $\theta = 0$ receives $\alpha r > \alpha$ — overpaid relative to the type-specific disagreement payoff. This overpayment arises because W cannot tailor the offer to the realized type and must offer enough for the high type.

On the aggressive branch, W offers α . Type $\theta = 0$ accepts (α), type $\theta = 1$ rejects and receives αr from disagreement. These are the same type-specific payoffs as under majority. $\square$ $\square$

Remark 4 (Where the mechanism lives). *The difference between unanimity and majority at R2 is not in the voting rule per se. It is in the information structure of the offer. Under majority, H is excluded and receives $\alpha V(\theta)$ — a type-specific payoff determined by nature, not by any strategic interaction. Under unanimity, W must make a single offer to H without knowing θ . When W plays conservative, this single offer necessarily overpays the low type. The screening channel and the jump in $E[V_H]$ are consequences of this forced pooling in W 's offer.*

6.1.4 Numerical Examples (R2 under Unanimity, $N = 5$, $\mu = 0.5$)

Example 1: $r = 1.1$, $\alpha = 0.5$ (bilateral captures 50% of multilateral value; $\alpha < 1/r = 0.909$ ✓).

Screening cutoff: $\mu_s = 0.5 \times 0.1 / (1.1 - 0.5) = 0.05 / 0.6 \approx 0.083$.

Since $\mu = 0.5 > \mu_s$, W plays **conservative** (offers $y_H = \alpha r = 0.55$).

Payoff matrix for W as proposer (expected over θ , given $\mu = 0.5$):

	(a, a)	(a, r)	(r, a)	(r, r)
A	0.25	0.25	0	0
C	0.50	0.225	0.50	0.225

Under (a, a) : W prefers C ($0.50 > 0.25$). ✓ Consistent with $\mu > \mu_s$.

Equilibrium payoffs and budget check:

The expected pie is $V_e(0.5) = 1.05$.

H expects to receive: 0.65 (62% of expected pie).

- If $\theta = 0$: H receives $[1 + 4 \times 0.55] / 5 = 0.64$ (from a pie of 1.0)
- If $\theta = 1$: H receives $[1.1 \times 3] / 5 = 0.66$ (from a pie of 1.1)
- Average: $0.5 \times 0.66 + 0.5 \times 0.64 = 0.65$

Each W expects to receive: 0.10 (9.5% of expected pie).

- W only gains when proposer (prob $1/5$).

- As proposer, W keeps $V(\theta) - \alpha r = V(\theta) - 0.55$:
 - If $\theta = 0$: $1 - 0.55 = 0.45$
 - If $\theta = 1$: $1.1 - 0.55 = 0.55$
- Expected as proposer: $0.5 \times 0.45 + 0.5 \times 0.55 = 0.50$
- Times 1/5: 0.10

Budget check: $H(0.65) + 4 \times W(0.10) = 0.65 + 0.40 = 1.05 = V_e(0.5)$. ✓

Jump magnitude at μ_s : $4 \times 0.5 \times 0.1 \times (1 - 0.083)/5 = 0.037$.

Interpretation: r close to 1 means the states are similar, α high means H has strong bilateral alternatives. The screening cutoff μ_s is very low (8.3%): W almost always plays conservative. The jump is small.

Example 2: $r = 1.5$, $\alpha = 0.3$ (bilateral captures 30%; $\alpha < 1/r = 0.667$ ✓).

Screening cutoff: $\mu_s = 0.3 \times 0.5 / (1.5 - 0.3) = 0.15 / 1.2 = 0.125$.

Since $\mu = 0.5 > \mu_s$, W plays **conservative** (offers $y_H = \alpha r = 0.45$).

Payoff matrix for W as proposer (expected over θ , given $\mu = 0.5$):

	(a, a)	(a, r)	(r, a)	(r, r)
A	0.35	0.35	0	0
C	0.80	0.275	0.80	0.275

Under (a, a) : W prefers C ($0.80 > 0.35$). ✓

Equilibrium payoffs and budget check:

The expected pie is $V_e(0.5) = 1.25$.

H expects to receive: 0.61 (49% of expected pie).

- If $\theta = 0$: H receives $[1 + 4 \times 0.45]/5 = 0.56$ (from a pie of 1.0)
- If $\theta = 1$: H receives $[1.5 \times 2.2]/5 = 0.66$ (from a pie of 1.5)
- Average: $0.5 \times 0.66 + 0.5 \times 0.56 = 0.61$

Each W expects to receive: 0.16 (12.8% of expected pie).

- As proposer (prob 1/5), keeps $V(\theta) - 0.45$:
 - If $\theta = 0$: $1 - 0.45 = 0.55$

– If $\theta = 1$: $1.5 - 0.45 = 1.05$

- Expected as proposer: $0.5 \times 0.55 + 0.5 \times 1.05 = 0.80$
- Times 1/5: 0.16

Budget check: $H(0.61) + 4 \times W(0.16) = 0.61 + 0.64 = 1.25 = V_e(0.5)$. ✓

Jump magnitude at μ_s : $4 \times 0.3 \times 0.5 \times (1 - 0.125)/5 = 0.105$.

Comparison: In Example 1 (α high, r low), H captures 62% of the expected pie — strong bilateral alternatives give bargaining power. In Example 2 (α low, r high), H captures 49% — weaker outside option. But the jump is larger in Example 2 (0.105 vs 0.037): the screening channel is most powerful precisely when H is “weaker” in terms of outside option.

Scaling with N : the screening cutoff $\mu_s = \alpha(r - 1)/(r - \alpha)$ does not depend on N . The jump $(N - 1)\alpha(r - 1)(1 - \mu_s)/N$ converges to $\alpha(r - 1)(1 - \mu_s)$ as $N \rightarrow \infty$. Individual payoffs scale as $1/N$. The table below shows how equilibrium shares evolve.

Example 1 ($r = 1.1, \alpha = 0.5, \mu = 0.5, \mu_s \approx 0.083$):

N	V_W	$V_H(\theta = 0)$	$V_H(\theta = 1)$	$E[V_H]$	H %	W %	Jump	Budget
5	0.1000	0.6400	0.6600	0.6500	61.9%	9.5%	0.0367	1.05
51	0.0098	0.5588	0.5608	0.5598	53.3%	0.9%	0.0449	1.05
101	0.0050	0.5545	0.5554	0.5550	52.9%	0.5%	0.0454	1.05
151	0.0033	0.5530	0.5536	0.5533	52.7%	0.3%	0.0455	1.05
201	0.0025	0.5522	0.5527	0.5525	52.6%	0.2%	0.0456	1.05

Example 2 ($r = 1.5, \alpha = 0.3, \mu = 0.5, \mu_s = 0.125$):

N	V_W	$V_H(\theta = 0)$	$V_H(\theta = 1)$	$E[V_H]$	H %	W %	Jump	Budget
5	0.1600	0.5600	0.6600	0.6100	48.8%	12.8%	0.1050	1.25
51	0.0157	0.4608	0.4706	0.4657	37.3%	1.3%	0.1287	1.25
101	0.0079	0.4554	0.4604	0.4579	36.6%	0.6%	0.1300	1.25
151	0.0053	0.4536	0.4570	0.4553	36.4%	0.4%	0.1304	1.25
201	0.0040	0.4527	0.4552	0.4540	36.3%	0.3%	0.1306	1.25

As N grows, each player’s individual payoff shrinks ($\sim 1/N$), but H ’s *share* of the pie converges to α (the bilateral fraction) and each W ’s share vanishes. The jump converges to $\alpha(r - 1)(1 - \mu_s)$, a positive constant independent of N . The screening channel does not disappear with large N .

6.2 Terminal Round (R2) under Majority

When H proposes (prob $1/N$): buys $q - 1$ votes (offers 0, $d_W = 0$). Keeps $V(\theta)$.

When W proposes (prob $(N - 1)/N$): W is indifferent between excluding H (coalition gets $(1 - \alpha)V(\theta)$, H captures $\alpha V(\theta)$ bilaterally) and including H at reservation $\alpha V(\theta)$ (W keeps $(1 - \alpha)V(\theta)$, H gets $\alpha V(\theta)$ from the deal). Payoffs are identical. **Convention:** W includes H in the proposal (WLOG). This ensures the full pie $V(\theta)$ is divided among all N players under both voting rules, simplifying comparison. W offers $\alpha V(\theta)$ to H , offers 0 to $q - 2$ other W 's in the coalition, and keeps $(1 - \alpha)V(\theta)$.

Note: under majority, the proposal passes regardless of H 's vote (W has $q - 1$ other votes). H accepts the offer $\alpha V(\theta)$ (weakly: it equals H 's outside option). There is no screening because H 's acceptance has no consequence for whether the deal passes — unlike unanimity, where H 's rejection kills the proposal.

$$V_H^{R2}(\theta, M) = \frac{V(\theta)[1 + (N - 1)\alpha]}{N} \quad \text{— linear in } \mu, \text{ no screening.} \quad (10)$$

$$V_W^{R2}(\mu, M) = \frac{(1 - \alpha)V_e(\mu)}{N} \quad (11)$$

Verification of $V_H^{R2}(\theta, M)$: H proposes with prob $1/N$ and keeps $V(\theta)$. W proposes with prob $(N - 1)/N$ and offers H exactly $\alpha V(\theta)$. So $V_H = (1/N)V(\theta) + ((N - 1)/N)\alpha V(\theta) = V(\theta)[1 + (N - 1)\alpha]/N$. Taking expectations: $E_\theta[V_H^{R2}(\mu, M)] = V_e(\mu)[1 + (N - 1)\alpha]/N$, affine in μ . ✓

Verification of $V_W^{R2}(\mu, M)$: W_i proposes with prob $1/N$ and keeps $(1 - \alpha)V(\theta)$ (offers $\alpha V(\theta)$ to H , 0 to other W 's). When H proposes (prob $1/N$), W_i receives 0. When another W_j proposes (prob $(N - 2)/N$), W_i receives 0. So $V_W = (1/N)(1 - \alpha)E[V(\theta)] = (1 - \alpha)V_e(\mu)/N$. ✓

Budget check: $E[V_H] + (N - 1)V_W = V_e(\mu)[1 + (N - 1)\alpha]/N + (N - 1)(1 - \alpha)V_e(\mu)/N = V_e(\mu) \cdot N/N = V_e(\mu)$. ✓

6.2.1 Numerical Examples and Comparison (R2, Majority vs Unanimity, $\mu = 0.5$)

Under majority there is no screening: W excludes H , so H always receives $\alpha V(\theta)$ when W proposes (type-specific, no overpayment).

Example 1 ($r = 1.1$, $\alpha = 0.5$, $\mu = 0.5$):

Majority:

N	V_W	$V_H(\theta=0)$	$V_H(\theta=1)$	$E[V_H]$	$H \%$	$W \%$	Budget
5	0.1050	0.6000	0.6600	0.6300	60%	10%	1.05
51	0.0103	0.5098	0.5608	0.5353	51%	1%	1.05
101	0.0052	0.5050	0.5554	0.5302	50.5%	0.5%	1.05
151	0.0035	0.5033	0.5536	0.5285	50.3%	0.3%	1.05
201	0.0026	0.5025	0.5527	0.5276	50.2%	0.2%	1.05

Unanimity (conservative branch):

N	V_W	$V_H(\theta=0)$	$V_H(\theta=1)$	$E[V_H]$	$H \%$	$W \%$	Jump	Budget
5	0.1000	0.6400	0.6600	0.6500	61.9%	9.5%	0.0367	1.05
51	0.0098	0.5588	0.5608	0.5598	53.3%	0.9%	0.0449	1.05
101	0.0050	0.5545	0.5554	0.5550	52.9%	0.5%	0.0454	1.05
151	0.0033	0.5530	0.5536	0.5533	52.7%	0.3%	0.0455	1.05
201	0.0025	0.5522	0.5527	0.5525	52.6%	0.2%	0.0456	1.05

Example 2 ($r = 1.5, \alpha = 0.3, \mu = 0.5$):

Majority:

N	V_W	$V_H(\theta=0)$	$V_H(\theta=1)$	$E[V_H]$	$H \%$	$W \%$	Budget
5	0.1750	0.4400	0.6600	0.5500	44%	14%	1.25
51	0.0172	0.3137	0.4706	0.3922	31.4%	1.4%	1.25
101	0.0087	0.3069	0.4604	0.3837	30.7%	0.7%	1.25
151	0.0058	0.3046	0.4570	0.3808	30.5%	0.5%	1.25
201	0.0044	0.3035	0.4552	0.3794	30.3%	0.3%	1.25

Unanimity (conservative branch):

N	V_W	$V_H(\theta=0)$	$V_H(\theta=1)$	$E[V_H]$	$H \%$	$W \%$	Jump	Budget
5	0.1600	0.5600	0.6600	0.6100	48.8%	12.8%	0.1050	1.25
51	0.0157	0.4608	0.4706	0.4657	37.3%	1.3%	0.1287	1.25
101	0.0079	0.4554	0.4604	0.4579	36.6%	0.6%	0.1300	1.25
151	0.0053	0.4536	0.4570	0.4553	36.4%	0.4%	0.1304	1.25
201	0.0040	0.4527	0.4552	0.4540	36.3%	0.3%	0.1306	1.25

6.2.2 H's gain from unanimity over majority

The gain $\Delta(\mu) = E_\theta[V_H^{R2}(U, \mu)] - E_\theta[V_H^{R2}(M, \mu)]$ has a simple closed form:

$$\Delta(\mu) = \begin{cases} 0 & \mu \leq \mu_s \\ \frac{(1 - \mu)(N - 1)\alpha(r - 1)}{N} & \mu > \mu_s \end{cases}$$

On the aggressive branch, payoffs are identical across rules. On the conservative branch, the gain comes entirely from the overpayment of $\theta = 0$ (Proposition 1).

Comparative statics (for $\mu > \mu_s$):

- $\partial\Delta/\partial\mu = -(N - 1)\alpha(r - 1)/N < 0$: gain is **maximized at** μ_s (just above the cutoff) and decreasing.
- $\partial\Delta/\partial r = (1 - \mu)(N - 1)\alpha/N > 0$: greater asymmetry between states $\Rightarrow$ larger gain.
- $\partial\Delta/\partial\alpha = (1 - \mu)(N - 1)(r - 1)/N > 0$: stronger bilateral alternatives $\Rightarrow$ larger gain at a given μ .
- $\Delta(\mu_s) = \text{Jump} = (N - 1)\alpha(r - 1)(1 - \mu_s)/N$: the maximum gain equals the jump magnitude.
- As $N \rightarrow \infty$: $\Delta \rightarrow \alpha(r - 1)(1 - \mu)$, a positive constant independent of N .

Note: $\partial\Delta/\partial\alpha > 0$ at a given μ does not mean the *maximum* gain is increasing in α . Higher α raises μ_s (shrinking the conservative region where the gain exists). The maximum gain $\Delta(\mu_s)$ depends on α through both the numerator and $(1 - \mu_s)$.

Example 1 ($r = 1.1$, $\alpha = 0.5$, $N = 5$, $\mu_s \approx 0.083$):

μ	Δ	% of pie	Max Δ	Max %
0.05	0.000	0%	0.0367	3.6%
0.10	0.036	3.6%	0.0367	3.6%
0.20	0.032	3.1%	0.0367	3.6%
0.30	0.028	2.7%	0.0367	3.6%
0.50	0.020	1.9%	0.0367	3.6%
0.70	0.012	1.1%	0.0367	3.6%
0.90	0.004	0.4%	0.0367	3.6%

Example 2 ($r = 1.5$, $\alpha = 0.3$, $N = 5$, $\mu_s = 0.125$):

μ	Δ	% of pie	Max Δ	Max %
0.05	0.000	0%	0.105	9.9%
0.10	0.000	0%	0.105	9.9%
0.20	0.096	8.7%	0.105	9.9%
0.30	0.084	7.3%	0.105	9.9%
0.50	0.060	4.8%	0.105	9.9%
0.70	0.036	2.7%	0.105	9.9%
0.90	0.012	0.8%	0.105	9.9%

The gain is modest in absolute terms but persistent: it does not vanish with N and is largest at low μ (just above μ_s), which is precisely the region where BP operates.

Key observations from the comparison:

- $V_H(\theta = 1)$ is **identical** across rules (always $r[1 + (N - 1)\alpha]/N$).
- $V_H(\theta = 0)$ is **higher under unanimity** (conservative): $[1 + (N - 1)\alpha r]/N > [1 + (N - 1)\alpha]/N$. This is the overpayment (Proposition 1).
- V_W is **higher under majority**: $V_e(\mu)/N > [V_e(\mu) - \alpha r]/N$. W keeps the whole pie under majority (excludes H) vs sharing with H under unanimity.
- As $N \rightarrow \infty$: H 's share converges to α under majority and to αr under unanimity (conservative). W 's share vanishes under both.
- The jump under unanimity converges to $\alpha(r - 1)(1 - \mu_s) > 0$, independent of N .

6.3 Round 1 (R1) under Unanimity

Let $x \equiv (N - 1)\alpha r$ throughout this section.

Remark 5 (Where screening lives). *On the equilibrium path, R2 is reached only when W plays aggressive in R1 and $\theta = 1$ rejects. In that case, W knows $\theta = 1$ with certainty ($\mu' = 1$). There is no screening in R2 on the equilibrium path. The screening of the model occurs in R1: W proposer chooses between aggressive and conservative offers to H without knowing θ . R2 serves as the off-path threat whose continuation values determine R1 reservation values.*

6.3.1 When H proposes in R1 (prob $1/N$)

H offers $\beta V_W^{R2}(\mu)$ to each W (their discounted R2 continuation value). Both types of H make the same offer (pooling: the offer depends on μ , not on θ , since V_W^{R2} is a function of μ only). All W 's accept (indifferent). H keeps $V(\theta) - (N - 1)\beta V_W^{R2}(\mu)$.

No information about θ is revealed by H 's proposal (same offer for both types).

6.3.2 When W proposes in R1 (prob $(N - 1)/N$)

W offers 0 to other W 's (their reservation from R2 is $\beta V_W^{R2}(\mu) > 0 \dots$ but other W 's accept 0 only if $0 \geq$ their continuation from rejection).

Note on other W 's offers: W_j must offer each W_k at least $\beta V_W^{R2}(\mu')$ where μ' depends on rejection beliefs. On the equilibrium path where the proposal is accepted, the exact value matters only for budget accounting. We set $x_k = \beta V_W^{R2}(\mu)$ for each W_k (using on-path beliefs). All W_k accept.

W_j must decide what to offer H . The two relevant offers are:

Conservative offer: $y_H^{con} = \beta(r + x)/N$

This is $\beta \times V_H^{R2}(\theta = 1)$, the discounted R2 continuation of the more demanding type. Both types accept:

- $\theta = 1$: $\beta(r + x)/N \geq \beta(r + x)/N$. Indifferent, accepts by convention.
- $\theta = 0$: $\beta(r + x)/N > \beta(1 + x)/N$. Strictly prefers to accept.

Game ends in R1.

Aggressive offer: $y_H^{agg} = \beta(1 + x)/N$

This is $\beta \times V_H^{R2}(\theta = 0, \mu' = 1)$, the discounted R2 continuation of type $\theta = 0$ under the belief $\mu' = 1$ (which is what W would believe after rejection, since only $\theta = 1$ rejects on path).

- $\theta = 0$: if $\theta = 0$ accepts, gets $\beta(1 + x)/N$. If $\theta = 0$ deviates and rejects, W believes $\theta = 1$ ($\mu' = 1$). In R2 with $\mu' = 1$, W offers αr to H (knows $\theta = 1$). Type $\theta = 0$ accepts αr (since $\alpha r > \alpha$). Full R2 payoff for $\theta = 0$: $(1 + x)/N$. Discounted: $\beta(1 + x)/N$. Exactly equal to the aggressive offer. **Indifferent, accepts by convention.**
- $\theta = 1$: $\beta(1 + x)/N < \beta(r + x)/N$ (since $1 < r$). **Strictly rejects.** Game goes to R2 with $\mu' = 1$ (complete information).

Gap: $\beta(r + x)/N - \beta(1 + x)/N = \beta(r - 1)/N > 0$. Screening is feasible.

6.3.3 R2 after R1 rejection (complete information)

When $\theta = 1$ rejects the aggressive offer in R1, the game goes to R2 with W knowing $\theta = 1$:

- H proposes (prob $1/N$): keeps $V(1) = r$.
- W proposes (prob $(N-1)/N$): offers αr to H (exact reservation under $\theta = 1$). Keeps $r - \alpha r = r(1 - \alpha)$.

R2 payoffs (under complete information, $\theta = 1$):

$$V_H^{R2}(\theta = 1, \text{known}) = \frac{r + x}{N} \quad (12)$$

$$V_W^{R2}(\theta = 1, \text{known}) = \frac{r(1 - \alpha)}{N} \quad (13)$$

6.3.4 W 's expected payoff from each strategy in R1

Let $\omega(\mu) \equiv (N-2)\beta V_W^{R2}(\mu)$ denote the total payment to other W 's in R1.

Conservative (both types accept, game ends in R1):

$$\begin{aligned} F_1^{con}(\mu) &= E[V(\theta)] - \frac{\beta(r + x)}{N} - \omega(\mu) \\ &= V_e(\mu) - \frac{\beta(r + x)}{N} - \omega(\mu) \end{aligned} \quad (14)$$

Aggressive ($\theta = 0$ accepts in R1; $\theta = 1$ rejects, goes to R2):

$$F_1^{agg}(\mu) = (1 - \mu) \left[V(0) - \frac{\beta(1 + x)}{N} - \omega(\mu) \right] + \mu \cdot \frac{\beta r(1 - \alpha)}{N} \quad (15)$$

The first term: with prob $(1 - \mu)$, $\theta = 0$ accepts. W keeps $V(0) - y_H^{agg} - \omega(\mu) = 1 - \beta(1 + x)/N - \omega(\mu)$.

The second term: with prob μ , $\theta = 1$ rejects. Game goes to R2 with complete information. W 's expected R2 payoff (discounted): $\beta V_W^{R2}(\theta = 1, \text{known}) = \beta r(1 - \alpha)/N$.

6.3.5 R1 screening cutoff

W prefers aggressive when $F_1^{agg}(\mu) > F_1^{con}(\mu)$:

$$(1 - \mu) \left[1 - \frac{\beta(1 + x)}{N} - \omega \right] + \frac{\mu\beta r(1 - \alpha)}{N} > V_e(\mu) - \frac{\beta(r + x)}{N} - \omega$$

Expanding and simplifying (the $\omega(\mu)$ terms partially cancel):

Left side:

$$(1 - \mu) - \frac{(1 - \mu)\beta(1 + x)}{N} - (1 - \mu)\omega + \frac{\mu\beta r(1 - \alpha)}{N}$$

Right side:

$$V_e(\mu) - \frac{\beta(r + x)}{N} - \omega$$

Difference (Left – Right):

$$(1 - \mu) - V_e(\mu) - \frac{(1 - \mu)\beta(1 + x)}{N} + \frac{\beta(r + x)}{N} + \frac{\mu\beta r(1 - \alpha)}{N} + \mu\omega$$

Note: $V_e(\mu) = 1 + \mu(r - 1)$, so $(1 - \mu) - V_e(\mu) = -\mu r$.

$$= -\mu r + \frac{\beta}{N} [-(1 - \mu)(1 + x) + (r + x) + \mu r(1 - \alpha)] + \mu\omega$$

The bracket:

$$\begin{aligned} & -(1 + x) + \mu(1 + x) + r + x + \mu r(1 - \alpha) \\ &= r - 1 + \mu(1 + x + r - r\alpha) \\ &= (r - 1) + \mu(1 + r + x - r\alpha) \end{aligned}$$

Since $x = (N - 1)\alpha r$, we have $x - r\alpha = (N - 2)\alpha r$. So:

$$= (r - 1) + \mu(1 + r + (N - 2)\alpha r)$$

The full difference:

$$\Delta_1(\mu) = -\mu r + \frac{\beta}{N} [(r-1) + \mu(1+r+(N-2)\alpha r)] + \mu\omega(\mu)$$

This depends on $\omega(\mu) = (N-2)\beta V_W^{R2}(\mu)$, which is piecewise (depends on which branch of V_W^{R2} applies). The cutoff μ_s^{R1} solves $\Delta_1(\mu) = 0$.

Rather than solving in full generality (the algebra is involved because V_W^{R2} is piecewise), we proceed numerically for the examples and note the closed-form structure.

6.3.6 Continuation values in R1 under unanimity

Averaging over proposer identity:

$V_H^{R1}(\theta, \mu)$:

$$\begin{aligned} &= \frac{1}{N} [V(\theta) - (N-1)\beta V_W^{R2}(\mu)] \quad (\text{H proposes}) \\ &+ \frac{N-1}{N} \times \begin{cases} \frac{\beta(1+x)}{N} & \text{if } \mu < \mu_s^{R1}, \theta = 0 \text{ (aggressive, accepts)} \\ \frac{\beta(r+x)}{N} & \text{if } \mu > \mu_s^{R1} \text{ (conservative, both accept)} \\ \beta V_H^{R2}(\theta = 1, \text{known}) = \frac{\beta(r+x)}{N} & \text{if } \mu < \mu_s^{R1}, \theta = 1 \text{ (aggressive, rejects } \rightarrow \text{R2)} \end{cases} \end{aligned} \quad (16)$$

Simplifying for each type:

Type $\theta = 1$:

$$V_H^{R1}(\theta = 1, \mu) = \frac{1}{N} [r - (N-1)\beta V_W^{R2}(\mu)] + \frac{(N-1)\beta(r+x)}{N^2} \quad (17)$$

This does not depend on whether W plays aggressive or conservative (type $\theta = 1$ gets $\beta(r+x)/N$ either way — accepts conservative, rejects aggressive and gets R2 payoff $(r+x)/N$). Constant across screening branches.

Type $\theta = 0$, aggressive ($\mu < \mu_s^{R1}$):

$$V_H^{R1}(\theta = 0, \mu < \mu_s^{R1}) = \frac{1}{N} [1 - (N-1)\beta V_W^{R2}(\mu)] + \frac{(N-1)\beta(1+x)}{N^2} \quad (18)$$

Type $\theta = 0$, conservative ($\mu > \mu_s^{R1}$):

$$V_H^{R1}(\theta = 0, \mu > \mu_s^{R1}) = \frac{1}{N} [1 - (N - 1)\beta V_W^{R2}(\mu)] + \frac{(N - 1)\beta(r + x)}{N^2} \quad (19)$$

Jump in $V_H^{R1}(\theta = 0)$ at μ_s^{R1} :

$$\text{Jump}_{\theta=0} = \frac{(N - 1)\beta(r - 1)}{N^2}$$

This comes from $\theta = 0$ receiving $\beta(r + x)/N$ (conservative) vs $\beta(1 + x)/N$ (aggressive) when W proposes, weighted by prob $(N - 1)/N$ of W proposing.

Jump in $E_\theta[V_H^{R1}(\mu)]$ at μ_s^{R1} :

$$\text{Jump} = (1 - \mu_s^{R1}) \times \frac{(N - 1)\beta(r - 1)}{N^2}$$

(Only $\theta = 0$ is affected; $\theta = 1$'s payoff is the same on both sides.)

6.3.7 Numerical examples (R1 under Unanimity)

Example 1: $r = 1.1, \alpha = 0.5, N = 5, \beta = 0.9$

$$\mu_s^{R2} = 0.083, x = (N - 1)\alpha r = 2.2$$

μ	Strat.	F_1^{agg}	F_1^{con}	$V_H^{R1}(0)$	$V_H^{R1}(1)$	$E[V_H^{R1}]$	V_W^{R1}	Jump	Budget
0.05	Aggressive	0.1641	0.1545	0.5924	0.6268	0.5941	0.0328	0.0137	0.7254
0.10	Aggressive	0.1679	0.1676	0.5946	0.6290	0.5980	0.0336	0.0130	0.7324
0.20	Conservative	0.1560	0.1722	0.6075	0.6275	0.6115	0.0344	0.0115	0.7493
0.30	Conservative	0.1451	0.1768	0.6061	0.6261	0.6121	0.0354	0.0101	0.7535
0.50	Conservative	0.1265	0.1860	0.6032	0.6232	0.6132	0.0372	0.0072	0.7620
0.70	Conservative	0.1123	0.1952	0.6003	0.6203	0.6143	0.0390	0.0043	0.7705
0.90	Conservative	0.1023	0.2044	0.5974	0.6174	0.6154	0.0409	0.0014	0.7790

Example 2: $r = 1.5, \alpha = 0.3, N = 5, \beta = 0.9$

$$\mu_s^{R2} = 0.125, x = (N - 1)\alpha r = 1.8$$

μ	Strat.	F_1^{agg}	F_1^{con}	$V_H^{R1}(0)$	$V_H^{R1}(1)$	$E[V_H^{R1}]$	V_W^{R1}	Jump	Budget
0.05	Aggressive	0.1395	0.0719	0.5074	0.6794	0.5160	0.0279	0.0684	0.6276
0.10	Aggressive	0.1591	0.1158	0.5125	0.6845	0.5297	0.0318	0.0648	0.6570
0.20	Conservative	0.1538	0.1550	0.5816	0.6816	0.6016	0.0310	0.0576	0.7256
0.30	Conservative	0.1393	0.1780	0.5744	0.6744	0.6044	0.0356	0.0504	0.7468
0.50	Conservative	0.1265	0.2240	0.5600	0.6600	0.6100	0.0448	0.0360	0.7892
0.70	Conservative	0.1353	0.2700	0.5456	0.6456	0.6156	0.0540	0.0216	0.8316
0.90	Conservative	0.1657	0.3160	0.5312	0.6312	0.6212	0.0632	0.0072	0.8740

6.4 Round 1 (R1) under Majority

Under majority, W includes H at reservation $\alpha V(\theta)$ (WLOG convention). The proposal passes regardless of H 's vote. No screening.

When H proposes (prob $1/N$): H offers $\beta V_W^{R2}(\mu, M) = \beta(1 - \alpha)V_e(\mu)/N$ to each W in the coalition. W accepts. H keeps $V(\theta) - (q - 1)\beta(1 - \alpha)V_e(\mu)/N$.

When W proposes (prob $(N - 1)/N$): W offers $\alpha V(\theta)$ to H (reservation, type-specific). But W doesn't know θ . Under majority, this doesn't matter: the proposal passes regardless. W offers $\beta\alpha V_e(\mu)$ (expected reservation) to H and keeps the rest. Actually, since the proposal passes by majority without H 's vote, W can offer any amount. By the WLOG convention (include H), W offers $\alpha V(\theta)$. Since W doesn't know $V(\theta)$, W offers $\beta \cdot \alpha V_e(\mu)$ and H gets $\alpha V(\theta)$ regardless (either from the deal or outside option).

In practice, the R1 payoffs under majority are just discounted R2 values:

$$V_H^{R1}(\theta, M) = \frac{V(\theta)[1 + (N - 1)\alpha \cdot \phi(\beta)]}{N} \quad (20)$$

$$V_W^{R1}(\mu, M) = \frac{(1 - \alpha)V_e(\mu) \cdot \psi(\beta)}{N} \quad (21)$$

where ϕ and ψ are functions of β from the standard BF backward induction. The key point: everything remains **linear in μ , no screening, no jump**.

A full derivation of R1 under majority parallels the unanimity case but without the aggressive/conservative distinction. Since there is no screening, the continuation values are affine in μ and do not contribute to BP. We defer the exact expressions to focus on the unanimity mechanism.

6.5 Entry

A weak state enters if its expected payoff from participating exceeds the entry cost:

$$V_W^{R1}(\mu, R) \geq c$$

The entry threshold $\tau(R)$ is the value of μ at which W is indifferent.

6.5.1 Entry under unanimity

$V_W^{R1}(\mu, U)$ is piecewise: it depends on whether W plays aggressive or conservative in R1 (cutoff μ_s^{R1}), and within the aggressive branch, on $V_W^{R2}(\mu)$ (cutoff μ_s). So V_W^{R1} can have up to three regimes.

The entry threshold $\tau(U)$ solves $V_W^{R1}(\tau, U) = c$. Since V_W^{R1} is not monotone in general (it depends on the piecewise structure), the threshold may not be unique. We characterize it numerically for the examples.

With $c = 0.1$, $\beta = 0.9$:

	$\tau(U)$	$\tau(M)$	$V_W^{R1}(0.5, U)$	$V_W^{R1}(0.5, M)$
Example 1 ($r = 1.1, \alpha = 0.5$)	0.01	0	0.1092	0.1239
Example 2 ($r = 1.5, \alpha = 0.3$)	0	0	0.16	0.2065

6.5.2 Entry under majority

Under majority, $V_W^{R1}(\mu, M)$ is affine in μ (no screening, no piecewise structure). The entry threshold $\tau(M)$ solves $V_W^{R1}(\tau, M) = c$, which has a unique solution.

6.5.3 Entry comparison

6.6 Bayesian Persuasion

H chooses the institutional rule R at Stage 0 and the signal structure π at Stage 1 to maximize expected payoff. By the standard Bayesian Persuasion result (Kamenica and Gentzkow 2011), H 's optimal payoff under rule R equals the concavification of H 's value function evaluated at the prior p :

$$\Pi_H^*(R) = \text{cav } v(p, R)$$

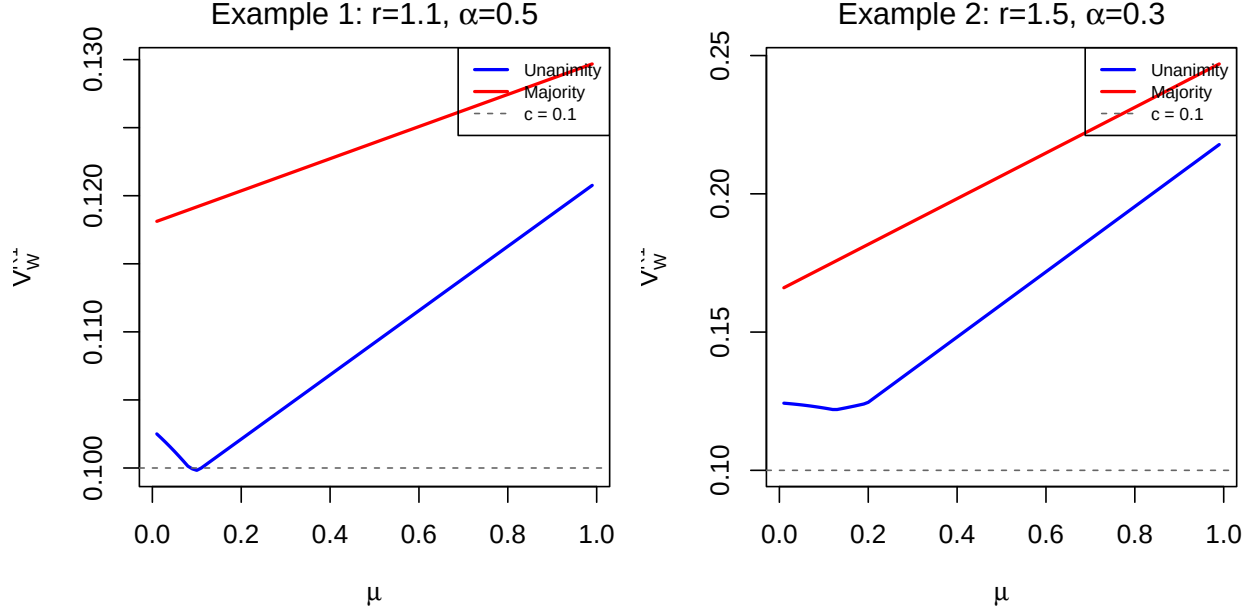


Figure 3: (#fig:entry_plot) $V_W^{R1}(\mu)$ under unanimity and majority for both examples. Horizontal line: entry cost $c = 0.1$. Entry occurs where the curve exceeds c .

6.6.1 H 's value function

H 's value function maps posteriors μ to H 's expected payoff:

$$v(\mu, R) = \begin{cases} 0 & \text{if } V_W^{R1}(\mu, R) < c \quad (\text{no entry}) \\ E_\theta[V_H^{R1}(\theta, \mu, R)] & \text{if } V_W^{R1}(\mu, R) \geq c \quad (\text{entry}) \end{cases}$$

When W does not enter, H gets 0 (no institution). When W enters, H gets the expected R1 payoff.

Under majority: $v(\mu, M) = \mathbf{1}\{V_W^{R1}(\mu, M) \geq c\} \cdot E_\theta[V_H^{R1}(\theta, \mu, M)]$. Both the indicator and the continuation value are affine (or piecewise affine) in μ . The value function has a jump at $\tau(M)$ (entry threshold) but is linear above it. BP exploits only the entry threshold.

Under unanimity: $v(\mu, U)$ has TWO sources of non-convexity:

1. **Entry threshold** at $\tau(U)$: v jumps from 0 to positive.
2. **Screening cutoff** at μ_s^{R1} : $E_\theta[V_H^{R1}]$ has an upward jump (from the overpayment of $\theta = 0$ when W switches from aggressive to conservative).

BP can exploit both non-convexities. This is the screening channel that is absent under majority.

6.6.2 Analytical concavification

Structure of $v(\mu, U)$. The value function under unanimity is:

$$v(\mu, U) = \begin{cases} 0 & \mu < \tau(U) \\ v_L(\mu) & \tau(U) \leq \mu < \mu_s^{R1} \\ v_H(\mu) & \mu \geq \mu_s^{R1} \end{cases}$$

where $v_L(\mu) = E_\theta[V_H^{R1}(\theta, \mu)]$ on the aggressive branch (pre-screening) and $v_H(\mu) = E_\theta[V_H^{R1}(\theta, \mu)]$ on the conservative branch (post-screening). There is an upward jump at μ_s^{R1} :

$$v_H(\mu_s^{R1}) - v_L(\mu_s^{R1}) = (1 - \mu_s^{R1}) \cdot \frac{(N-1)\beta(r-1)}{N^2} > 0.$$

Both v_L and v_H depend on $V_W^{R2}(\mu)$ and are continuous on their respective domains.

Structure of $v(\mu, M)$. Under majority:

$$v(\mu, M) = \begin{cases} 0 & \mu < \tau(M) \\ E_\theta[V_H^{R1}(\theta, \mu, M)] & \mu \geq \tau(M) \end{cases}$$

This is affine in μ for $\mu \geq \tau(M)$. No jump, no non-convexity beyond the entry threshold.

Case 1: $\tau(U) > 0$, prior $p < \tau(U)$ (entry does not occur at the prior).

The concavification from the origin creates a line with slope:

$$S(R) = \sup_{\mu \in [\tau(R), 1]} \frac{v(\mu, R)}{\mu}$$

and $\text{cav } v(p, R) = S(R) \cdot p$ for $p < \tau(R)$.

Under majority: $v(\mu, M) = A_M + B_M \mu$ is affine for $\mu \geq \tau(M)$, so v/μ is decreasing. The supremum is at $\mu = \tau(M)$:

$$S_M = \frac{v(\tau(M), M)}{\tau(M)}$$

Under unanimity: v/μ can attain a higher value at the jump. If $v_H(\mu_s^{R1})/\mu_s^{R1} > v_L(\mu)/\mu$ for all

$\mu \in [\tau(U), \mu_s^{R1})$, the supremum is at μ_s^{R1} (from the right):

$$S_U = \frac{v_H(\mu_s^{R1})}{\mu_s^{R1}}$$

Proposition 2 (BP dominance at low priors). *If $S_U > S_M$, then H strictly prefers unanimity for all priors $p \in (0, \min(\tau(U), \tau(M)))$:*

$$\text{cav } v(p, U) = S_U \cdot p > S_M \cdot p = \text{cav } v(p, M).$$

The condition $S_U > S_M$ is equivalent to:

$$\frac{v_H(\mu_s^{R1})}{\mu_s^{R1}} > \frac{v(\tau(M), M)}{\tau(M)}.$$

The left side is the “return per unit of belief” at the screening cutoff under unanimity. The right side is the return at the entry threshold under majority. Unanimity dominates when the screening jump creates a sufficiently high return relative to majority’s entry-threshold return.

Case 2: $\tau(U) \approx 0$, prior $p > \tau(U)$ (entry occurs at the prior).

In this case, $v(p, U) > 0$. The BP gain comes from exploiting the screening jump at μ_s^{R1} . The concavification bridges the non-convexity: for priors near μ_s^{R1} , H creates posteriors straddling the jump.

Formally, there exist $\mu_1 < \mu_s^{R1} < \mu_2$ such that the line connecting $(\mu_1, v_L(\mu_1))$ to $(\mu_2, v_H(\mu_2))$ is tangent to v_L at μ_1 and tangent to v_H at μ_2 . For $p \in (\mu_1, \mu_2)$:

$$\text{cav } v(p, U) = v_L(\mu_1) + \frac{v_H(\mu_2) - v_L(\mu_1)}{\mu_2 - \mu_1}(p - \mu_1) > v(p, U).$$

The optimal signal creates two posteriors: μ_1 (with prob $(\mu_2 - p)/(\mu_2 - \mu_1)$) and μ_2 (with prob $(p - \mu_1)/(\mu_2 - \mu_1)$). At μ_1 , W plays aggressive. At μ_2 , W plays conservative. H benefits from the switch.

Under majority, $v(\mu, M)$ is concave for $\mu \geq \tau(M)$, so $\text{cav } v(p, M) = v(p, M)$. No BP gain beyond entry.

Proposition 3 (BP gain from screening). *Under unanimity, $\text{cav } v(p, U) > v(p, U)$ for all $p \in (\mu_1, \mu_2)$, with BP gain:*

$$\text{cav } v(p, U) - v(p, U) > 0.$$

Under majority, $\text{cav } v(p, M) = v(p, M)$ for all $p \geq \tau(M)$: no BP gain beyond entry.

The screening channel is therefore the unique source of differential BP advantage for unanimity over majority.

Summary of the two channels:

1. **Entry channel:** if $\tau(U) < \tau(M)$, unanimity has a lower entry threshold, giving BP a longer “reach” at low priors. This channel exists regardless of screening.
2. **Screening channel:** the jump at μ_s^{R1} creates a non-convexity in $v(\mu, U)$ that BP exploits. This channel is absent under majority (no jump, v is concave). This is the paper’s distinctive contribution.

6.6.3 Numerical illustration

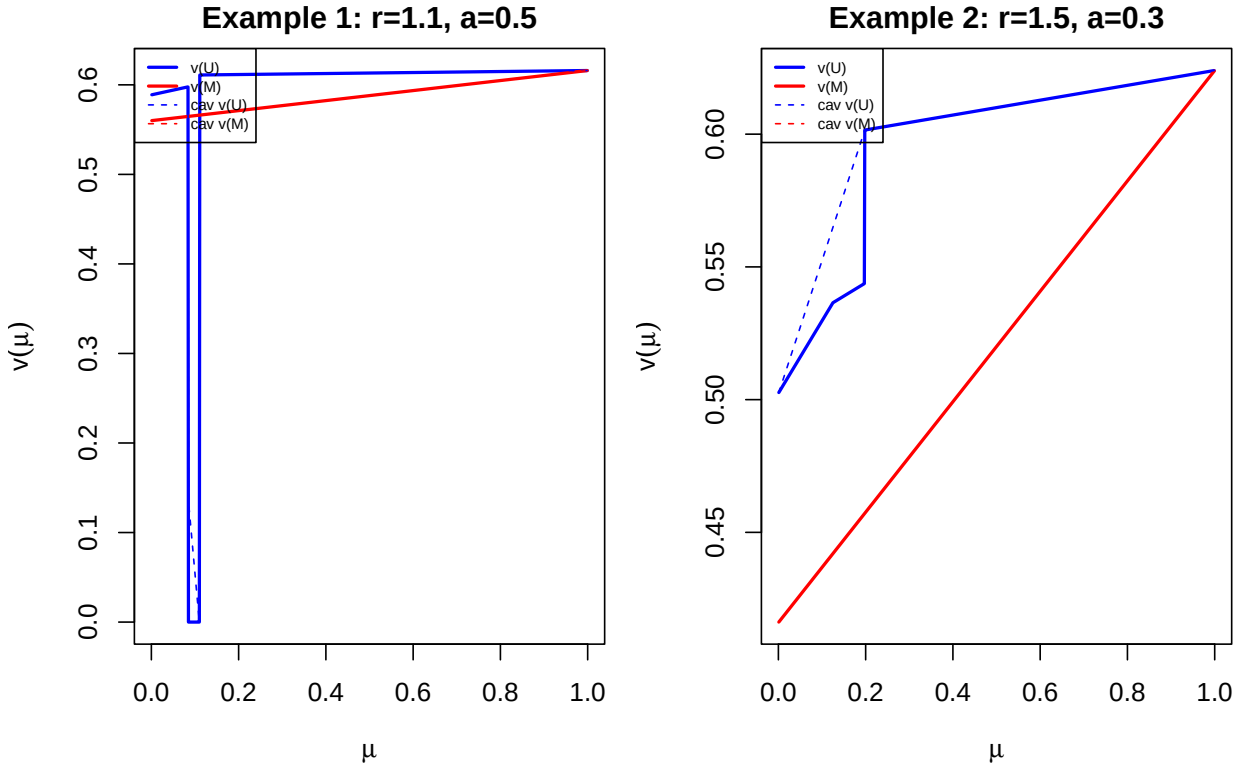


Figure 4: (#fig:bp_plot) H ’s value function $v(\mu, R)$ and its concavification under both rules. Dashed: concave envelope. The screening jump under unanimity creates additional concavification gain.

Key observation: The concavification under unanimity exploits both the entry threshold and the screening jump. Under majority, only the entry threshold is available. For priors p where $\text{cav } v(p, U) > \text{cav } v(p, M)$, H prefers unanimity.

6.6.4 H 's optimal payoff comparison

p	Example 1		Example 2	
	$\text{cav } v(p, U)$	$\text{cav } v(p, M)$	$\text{cav } v(p, U)$	$\text{cav } v(p, M)$
0.05	0.5941	0.5628	0.5273	0.4264
0.1	0.054	0.5656	0.5524	0.4368
0.15	0.6112	0.5684	0.5775	0.4472
0.2	0.6115	0.5712	0.6016	0.4576
0.25	0.6118	0.574	0.603	0.468
0.3	0.6121	0.5768	0.6044	0.4784
0.35	0.6124	0.5796	0.6058	0.4888
0.4	0.6126	0.5824	0.6072	0.4992
0.45	0.6129	0.5852	0.6086	0.5096
0.5	0.6132	0.588	0.61	0.52
0.55	0.6135	0.5908	0.6114	0.5304
0.6	0.6138	0.5936	0.6128	0.5408
0.65	0.614	0.5964	0.6142	0.5512
0.7	0.6143	0.5992	0.6156	0.5616
0.75	0.6146	0.602	0.617	0.572
0.8	0.6149	0.6048	0.6184	0.5824
0.85	0.6152	0.6076	0.6198	0.5928
0.9	0.6154	0.6104	0.6212	0.6032
0.95	0.6157	0.6132	0.6226	0.6136

6.7 Institutional Comparison without BP

Before deriving the full theorem with BP, we characterize the comparison *without* Bayesian Persuasion (i.e., W uses the prior p directly, no signal).

Proposition 4 (Comparison without BP). *Fix parameters (N, r, α, β, c) in the standard range ($r > 1$, $\alpha < 1/r$, β not too low).*

1. W always prefers majority: $V_W^{R1}(\mu, U) < V_W^{R1}(\mu, M)$ for all $\mu \in [0, 1]$.
2. Conditional on entry, H prefers unanimity: $E_\theta[V_H^{R1}(\theta, \mu, U)] > E_\theta[V_H^{R1}(\theta, \mu, M)]$ for all μ above both entry thresholds. The overpayment of $\theta = 0$ under unanimity dominates the coalition-size advantage under majority.
3. Entry is harder under unanimity: $\tau(U) \geq \tau(M)$ (since $V_W(U) < V_W(M)$, W requires a

higher μ to justify entry).

Remark 6 (Pareto ranking without BP). *The comparison depends on the prior p :*

Prior range	W prefers	H prefers	Pareto ranking
$p < \tau(M)$	indifferent	indifferent	both get 0
$\tau(M) < p < \tau(U)$	M	M	M Pareto dominates
$p > \tau(U)$	M	U	not comparable

For priors in $(\tau(M), \tau(U))$, entry occurs only under majority: both players prefer M (something vs nothing). Without BP, unanimity is **not** Pareto superior.

Remark 7 (The role of BP — revised). H already prefers unanimity conditional on entry (even without BP). The problem is that entry is harder under unanimity ($\tau(U) > \tau(M)$). BP solves this in two ways:

1. **Entry inducement:** BP can induce entry at priors $p < \tau(U)$ where W would not enter at the prior.
2. **Screening exploitation:** the jump in $v(\mu, U)$ at μ_s^{R1} creates a non-convexity that concavification exploits, giving H additional gain.

BP does not invert H 's preference (H already wants unanimity). BP viabilizes unanimity by making entry feasible at priors where it would otherwise fail.

Remark 8 (Parameter conditions). The ranking $V_W(U) < V_W(M)$ and $E[V_H(U)] > E[V_H(M)]$ (conditional on entry) holds for standard parameters ($\beta \geq 0.7$, $\alpha \leq 0.5$). For extreme parameters ($\beta \leq 0.5$ or $\alpha \geq 0.6$), the H comparison may reverse. The theorem should state the parametric conditions explicitly.

6.8 Institutional Comparison with BP — to derive (Theorem)

7 Open Questions

1. **Direction of entry without direct benefit:** under unanimity (aggressive branch), V_W is decreasing in μ — entry easier at low μ . Under majority, V_W is increasing — entry easier at high μ . Opposite directions. Derive consequences for BP.
2. **Feasibility of absolute offers:** W proposes amounts without knowing $V(\theta)$. Conservative offer $y_H = \alpha r$: feasible when $\theta = 0$ ($\alpha r < 1$ since $\alpha < 1/r$)? Yes. ✓
3. V_W **under majority:** when W proposes and excludes H , W keeps $V(\theta)$ but does not know

$V(\theta)$. W proposes amounts and receives residual $V(\theta) - \text{offers}$. Residual depends on θ .

4. **Continuity of V_W^{R2} at μ_s :** verify algebraically.
5. **Condition for H to prefer unanimity:** when does screening channel + superior extraction outweigh entry disadvantage?